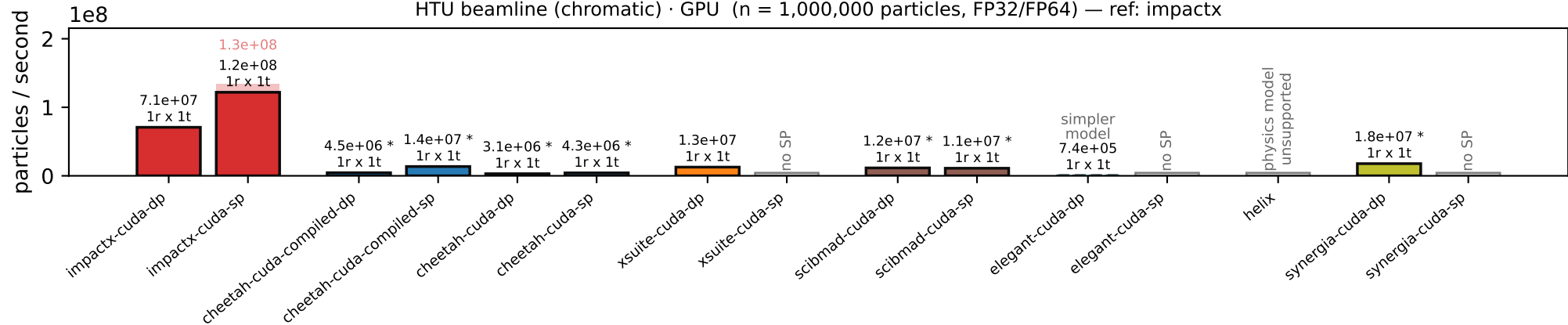


HTU beamline (chromatic) · GPU (n = 1,000,000 particles, FP32/FP64) — ref: impactx



'simpler model' = a lower-fidelity model vs the reference — elegant: tracks geometric slope (x,x') not canonical (x,px)

lighter bar behind + lighter value above = fast-math (relaxed FP), shown when $\geq 5\%$ faster than the IEEE bar

* Cheetah/SciBmad/Bmad/Synergia have no chromatic-paraxial model here; they run the costlier exact map (agrees within tol)

versions: impactx 26.08 · cheetah 0.8.2 · xsuite 0.53.4 · scibmad 0.4.1 (v0.4.1) · elegant elegant-2026.4.0 · helix v1.10.1 · synergia v2024.04-209-g795b99f39